

关于 IMX6U 核心板说明报告单

Emmc 对旧固件的兼容性说明

二〇二三年六月

一、问题简要说明

IMX6U 核心板使用旧版本固件出现 emmc 不能正常识别的情况
识别失败的情况:

```
[ 3.700496] sdhci: Tuning procedure failed, falling back to fixed sampling clock
[ 3.710357] mmc1: tuning execution failed
[ 3.714748] mmc1: error -5 whilst initialising MMC card
[ 3.721724] dhd_module_init in
[ 3.726305] input: gpio_keys@0 as /devices/platform/gpio_keys@0/input/input1
[ 3.734251] snvs_rtc 20cc000.snvs:snvs-rtc-lp: setting system clock to 2022-03-05 18:22:08 UTC (1646504528)
[ 3.760690] gpio_dvfs: disabling
[ 3.764055] can-3v3: disabling
[ 3.767394] ALSA device list:
[ 3.770377] No soundcards found.
[ 3.793611] EXT3-fs (mmcblk0p2): error: couldn't mount because of unsupported optional features (240)
[ 3.808297] EXT2-fs (mmcblk0p2): error: couldn't mount because of unsupported optional features (244)
[ 3.890861] mmc1: MAN_BKOPS_EN bit is not set
[ 3.943009] sdhci: Tuning procedure failed, falling back to fixed sampling clock
[ 3.952349] mmc1: tuning execution failed
[ 3.956474] mmc1: error -5 whilst initialising MMC card
[ 4.110628] mmc1: MAN_BKOPS_EN bit is not set
[ 4.155107] sdhci: Tuning procedure failed, falling back to fixed sampling clock
[ 4.162727] mmc1: tuning execution failed
[ 4.166902] mmc1: error -5 whilst initialising MMC card
[ 4.357529] mmc1: MAN_BKOPS_EN bit is not set
[ 4.363161] EXT4-fs (mmcblk0p2): recovery complete
[ 4.372849] EXT4-fs (mmcblk0p2): mounted filesystem with ordered data mode. Opts: (null)
[ 4.382110] VFS: Mounted root (ext4 filesystem) on device 179:2.
```

正常识别的情况:

```
3 [ 3.610743] mmc1: MAN_BKOPS_EN bit is not set
9 [ 3.615404] Bluetooth: RFCOMM socket layer initialized
0 [ 3.621723] Bluetooth: RFCOMM ver 1.11
1 [ 3.625783] Bluetooth: BNEP (Ethernet Emulation) ver 1.3
2 [ 3.632225] Bluetooth: BNEP filters: protocol multicast
3 [ 3.638706] Bluetooth: BNEP socket layer initialized
4 [ 3.643884] Bluetooth: HIDP (Human Interface Emulation) ver 1.2
5 [ 3.650939] Bluetooth: HIDP socket layer initialized
6 [ 3.660373] 8021q: 802.1Q VLAN Support v1.8
7 [ 3.667981] Key type dns_resolver registered
8 [ 3.703162] mmc1: new HS200 MMC card at address 0001
9 [ 3.710661] mmcblk1: mmc1:0001 SPeMMC 7.27 GiB
0 [ 3.716033] dhd_module_init in
1 [ 3.720452] input: gpio_keys@0 as /devices/platform/gpio_keys@0/input/input1
2 [ 3.727963] mmcblk1boot0: mmc1:0001 SPeMMC partition 1 4.00 MiB
3 [ 3.734899] snvs_rtc 20cc000.snvs:snvs-rtc-lp: setting system clock to 2019-08-17 02:09:52 UTC (1566007792)
4 [ 3.744804] mmcblk1boot1: mmc1:0001 SPeMMC partition 2 4.00 MiB
5 [ 3.751011] mmcblk1rpm: mmc1:0001 SPeMMC partition 3 4.00 MiB
6 [ 3.765739] mmcblk1: p1 p2
7 [ 3.781194] gpio_dvfs: disabling
8 [ 3.784641] can-3v3: disabling
9 [ 3.787982] ALSA device list:
0 [ 3.790967] No soundcards found.
1 [ 3.804184] EXT3-fs (mmcblk0p2): error: couldn't mount because of unsupported optional features (240)
```

二、问题分析

内核在初始化 emmc 驱动的时候出现异常报错, emmc 不能正常初始化使用。

三、解决措施

需要在设备树上添加一个属性字段，让 emmc 驱动可以正常初始化。

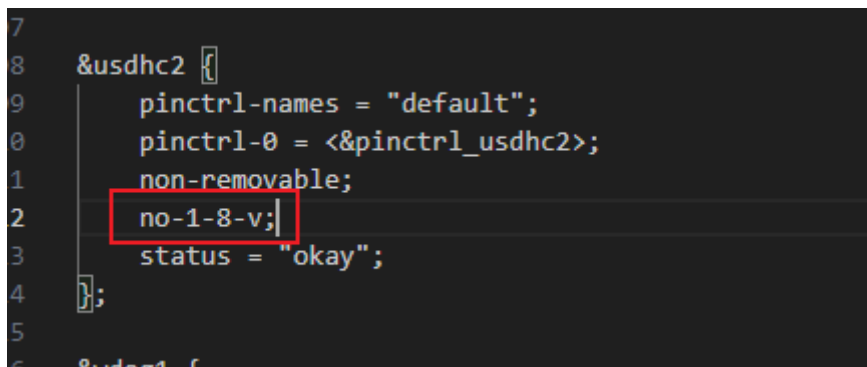
操作步骤如下：

打开内核源码，找到产品对应的设备树。下面基于我们正点原子的出厂内核源码为例，客户有自己定义的设备树请在自己的设备树上添加，原理一样。

正点原子 IMX6U 开发板使用的设备树位于内核源码目录下的：arch/arm/boot/dts/imx6ull-14x14-evk.dts

打开设备树找到 “usdhc2” 节点，没有的话可以在自己设备树末尾追加新节点在 “usdhc2” 节点里面添加 “no-1-8-v;” 属性字段。

具体如下图：



```
7
8  &usdhc2 {
9      pinctrl-names = "default";
10     pinctrl-0 = <&pinctrl_usdhc2>;
11     non-removable;
12     no-1-8-v;
13     status = "okay";
14 };
15
16 &wdog1 {
```

给 “usdhc2” 节点添加 “no-1-8-v;” 属性字段，即可解决这个问题

添加好之后，将文件保存，重新编译内核设备树。将新生成的设备树 dtb 文件替换到自己固件里面，重新烧写进去。